

CampusSphere

Feature Roadmap and Product Expansion Plan

AI-powered learning

Connected campus

Secure administration

Purpose

This roadmap lists the major features that can turn the current CampusSphere portal into a complete campus ERP, digital learning platform, and communication hub. The priorities are designed to help build in safe, useful stages.

Prepared for the next development phase

1. Foundation and Core Campus ERP

Build these first so every later module has reliable data, permissions, and communication.

Area	Features to add	Priority
Authentication	Forgot password, email verification, reset password, session management, profile photo, password change, optional two-factor login.	P0
Roles	Complete permissions for Super Admin, Admin, Principal, HOD, Faculty, Student, Placement Officer, Parent and Support Staff.	P0
College setup	College profile, campuses, departments, programs, batches, semesters, sections, academic calendar and holidays.	P0
Student records	Admission profile, documents, guardian details, roll number, enrollment status, fee category and student ID card.	P0
Faculty records	Employee profile, designation, department, workload, qualification, joining date and faculty ID card.	P0
Notifications	In-app alerts, email alerts, read/unread status, priority notices and scheduled announcements.	P0
Audit and safety	Admin audit trail, password-policy rules, login history, soft delete, backup plan and activity monitoring.	P0

Recommended data rule

Use one source of truth for users, departments, courses, subjects, enrollments and academic terms. Every module - attendance, marks, chat, meetings and reports - should use these shared records.

2. Learning, Assessment and AI

Features that make the platform valuable every day for students and faculty.

Area	Features to add	Priority
Course learning	Course catalogue, lesson modules, syllabus tracker, prerequisites, enrollment, progress bars and completion certificates.	P1
Study material	Folders by course and subject, video/PDF uploads, preview, search, bookmarks, download control and version history.	P1
Attendance	QR/manual attendance, timetable integration, late entry rule, absence alerts, subject-wise percentage and shortage notices.	P1
Exams	Question bank, online quizzes, timed tests, assignments, rubrics, answer upload, plagiarism check and result publishing.	P1
Performance	Marks dashboard, GPA/CGPA, subject analytics, weak-topic alerts, faculty feedback and improvement plans.	P1
AI chatbot	CampusSphere AI Assistant for FAQs, navigation help, syllabus/material Q&A;,, deadline reminders and role-aware answers.	P1
AI study coach	Personal study plan, quiz generation, flashcards, topic summaries, doubt explanation, revision schedule and progress nudges.	P2
AI faculty tools	Draft announcements, generate question papers, create lesson plans, summarize submissions and identify at-risk students.	P2

AI guardrails

- Keep chatbot answers role-aware: students must never see private faculty/admin data.
- Show source links for policies, notices and study material answers.
- Store AI conversation history with consent, reporting and delete controls.
- Use moderation and clear wording: AI suggestions are guidance, not official academic decisions.

3. Collaboration and Video Conferencing

Make CampusSphere the shared workspace for classes, mentoring and administration.

Area	Features to add	Priority
Messaging	One-to-one chat, course/section groups, faculty announcements, file sharing, message search, mentions and mute controls.	P1
Discussion forums	Subject-wise questions, answers, voting, moderation, tags, accepted answers and peer learning spaces.	P1
Appointments	Student-faculty booking slots, office hours, approval workflow, calendar sync and reminder notifications.	P1
Video meetings	Create/join meetings, waiting room, host controls, mic/camera controls, screen sharing, participant list and meeting chat.	P2
Live classes	Class scheduling, attendance from meeting, recording links, whiteboard, polls, raised hand, breakout rooms and captions.	P2
Recording library	Secure recordings by subject, permission-based playback, notes, timestamps, transcript and retention rules.	P2
Mentoring	Mentor matching, goals, meeting history, action items, feedback and escalation to HOD/counsellor.	P2

Video implementation choice

For a fast first version, integrate Jitsi Meet or Zoom/Google Meet links with scheduling and attendance. For an advanced in-app experience, use WebRTC with a managed provider such as Daily, Agora or Twilio. Start with secure meeting links, host controls and recordings before building complex live-class tools.

4. Operations, Engagement and Integrations

Area	Features to add	Priority
Fees	Fee structure, invoices, online payment gateway, dues reminders, receipts, scholarships and finance reports.	P2
Library	Book catalogue, issue/return, due dates, fines, e-books, seat booking and library analytics.	P2
Placement	Company drives, job posts, eligibility rules, resume bank, application tracking, interview schedule and offer analytics.	P2
Grievance	Confidential complaint tickets, category routing, SLA, status tracking, evidence upload and resolution reports.	P2
Events	College events, registrations, QR check-in, certificates, clubs, volunteer roles and photo gallery.	P2
Parent portal	Attendance, results, notices, fees, meeting booking and controlled communication with faculty.	P3
Mobile app	Android/iOS companion app, push notifications, QR attendance, offline downloads and quick chat access.	P3
Integrations	Google/Microsoft calendar, payment gateway, email/SMS/WhatsApp, LMS import/export, biometric device and SSO.	P3

Reports dashboard

- Admin: enrollment, attendance, fees, materials, user activity and campus-wide alerts.
- HOD: department performance, faculty workload, subject performance and pending approvals.
- Faculty: student progress, attendance shortages, assignment status and class engagement.
- Student: timetable, progress, deadlines, attendance, results and recommended next actions.

5. Recommended Development Sequence

Do not build everything at once. Deliver useful modules in small, testable releases.

Phase	Goal	Main deliverables
Phase 1	Stabilize core	Roles, permissions, profile fixes, notifications, academic setup, stronger user/admin management.
Phase 2	Daily academics	Timetable, attendance, materials, assessments, results and student/faculty dashboards.
Phase 3	AI learning	CampusSphere AI chatbot, content search, study plans, quizzes and faculty productivity tools.
Phase 4	Collaboration	Messaging, appointments, video meetings, live classes, recordings and mentoring.
Phase 5	Full ERP	Fees, library, placement, grievance, parent portal, mobile app and external integrations.

Success checklist

- Every feature has a clear role permission, mobile-friendly screen and empty/error state.
- Every sensitive action has confirmation, audit log and secure backend enforcement.
- Use real student/faculty feedback after each phase before building the next one.
- Add automated tests, database backups, environment-based configuration and deployment monitoring before production launch.